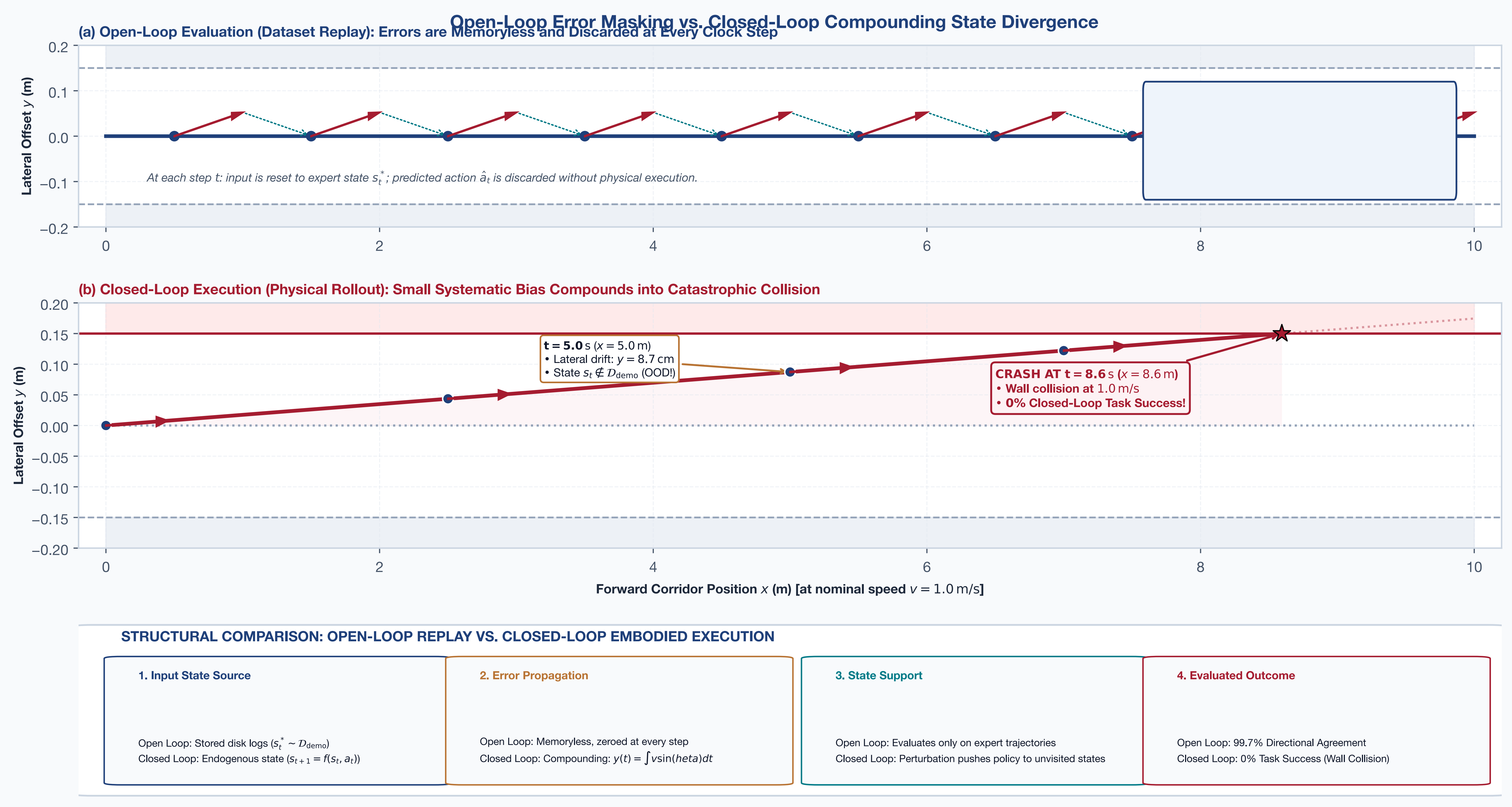

Open-Loop Error Masking vs. Closed-Loop Compounding State Divergence
(a) Open-Loop Evaluation (Dataset Replay): Errors are Memoryless and Discarded at Every Clock Step
Lateral Offset y (m)
At each step t: input is reset to expert state s_t^*; predicted action $\hat{a}_t$ is discarded without physical execution.
(b) Closed-Loop Execution (Physical Rollout): Small Systematic Bias Compounds into Catastrophic Collision
Lateral Offset y (m)
t = 5.0 s ($x = 5.0$ m)
• Lateral drift: $y = 8.7$ cm
• State $s_t \notin \mathcal{D}_{\mathrm{demo}}$ (OOD!)
CRASH AT t = 8.6 s ($x = 8.6$ m)
• Wall collision at 1.0 m/s
• 0% Closed-Loop Task Success!
Forward Corridor Position x (m) [at nominal speed $v = 1.0$ m/s]
STRUCTURAL COMPARISON: OPEN-LOOP REPLAY VS. CLOSED-LOOP EMBODIED EXECUTION
1. Input State Source
Open Loop: Stored disk logs ($s_t^* \sim \mathcal{D}_{\mathrm{demo}}$)
Closed Loop: Endogenous state ($s_{t+1} = f(s_t, a_t)$)
2. Error Propagation
Open Loop: Memoryless, zeroed at every step
Closed Loop: Compounding: $y(t) = \int v\sin(heta)dt$
3. State Support
Open Loop: Evaluates only on expert trajectories
Closed Loop: Perturbation pushes policy to unvisited states
4. Evaluated Outcome
Open Loop: 99.7% Directional Agreement
Closed Loop: 0% Task Success (Wall Collision)